CSE4088 Introduction to Machine Learning

Course Information

CSE4088 Intro. to Machine Learning

- Instructor: Çiğdem Eroğlu Erdem
- E-mail: cigdem.erdem@marmara.edu.tr

Prerequisites

- Calculus
- Probability and statistics
- Linear Algebra
- Programming (Python or MATLAB)

Course Objectives

- Undergraduate level introductory course in machine learning
- Give an overview of many concepts in machine learning:
  - theory of generalization
  - supervised learning methods
  - unsupervised learning methods.
- The students who successfully complete the course will be able to apply these concepts to real-world problems.

Learning Outcomes

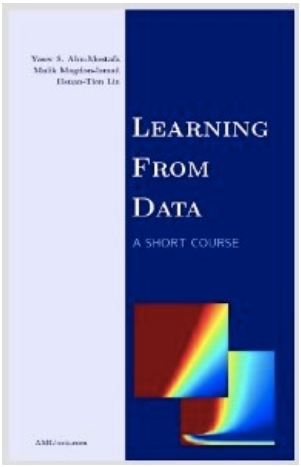
- **LO1:** Explain theoretical concepts of machine learning (theory of generalization, VC dimension etc.)
- **LO2:** Apply basic supervised classification methods (e.g. k-NN, decision trees, naive Bayes, support vector machines, neural networks).
- **LO3:** Apply basic regression (linear regression etc.) and unsupervised clustering (K-means etc.) algorithms in machine learning.
- **LO4:** Analyze features and apply feature selection and dimensionality reduction (PCA etc.) methods.

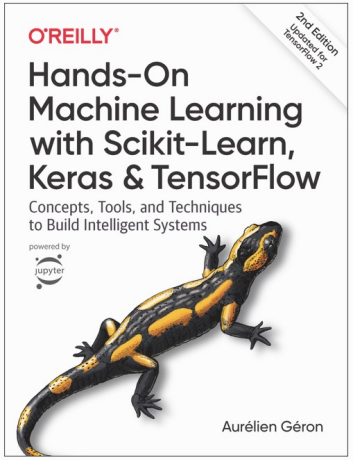
Learning Outcomes

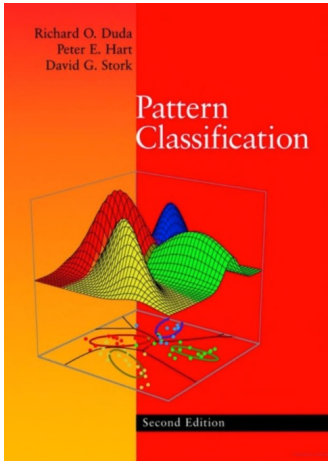
- **LO5:** Explain boosting and ensemble learning methods.
- **LO6:** Design experiments to evaluate and compare different machine learning techniques on real-world problems.
- **LO7:** Integrate multiple facets of practical machine learning in a single system: data preprocessing, learning, regularization and model selection

Reference Books

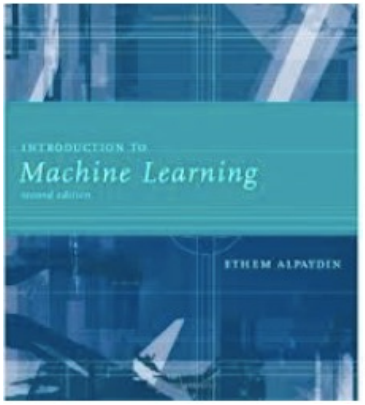
- Group 1:
  - **Learning from Data,** by Yaser S. Abu-Mostafa, Malik Magdon-Ismail, Hsuan-Tien Lin, 2012. (Book web page: http://work.caltech.edu/textbook.html)
  - **Hands-On Machine Learning with Scikit-Learn and TensorFlow**: Concepts, Tools, and Techniques to Build Intelligent Systems, Aurelien Geron, O'Reilly, 2017.
  - **A Course in Machine Learning**, Hal Daume III, 2017. (available online: https://ciml.info)
  - **Machine Learning,** Tom Mitchell. (http://www.cs.cmu.edu/afs/cs.cmu.edu/user/mitchell/ftp/mlbook.html)
  - **Pattern Classification, 2nd Edition,** R. O. Duda, P. E. Heart, D. G. Stork, Wiley, 2000.
  - **Deep Learning**, Ian GoodFellow, Yoshua Bengio, Aaron Courville, MIT Press, 2016. (available online: https://www.deeplearningbook.org/)







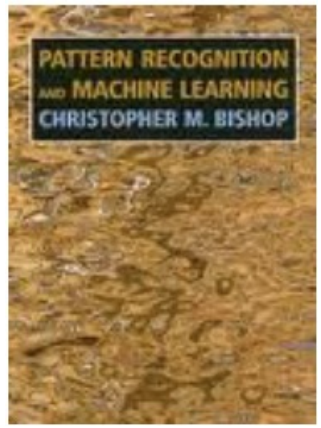


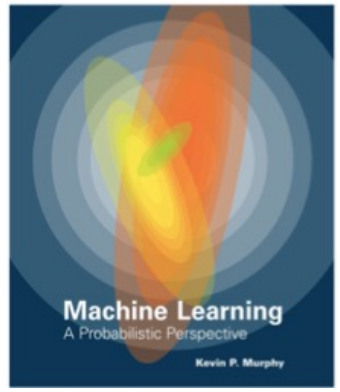




Reference Books

- Group 2:
  - **Introduction to Machine Learning with Python**, Andreas Muller and Sarah Guido, O'Reilly, 2017.
  - **Bayesian Reasoning and Machine Learning**, David Barber, Cambridge University Press, 2012. (available online: http://www.cs.ucl.ac.uk/staff/d.barber/brml/)
  - **Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow**, Sebastian Raschka and Vahid Mirjalili, 2nd Edition, Packt Publishing, 2017.
  - **Pattern Recognition and Machine Learning**, C. M. Bishop, Springer, 2013. (available online)
  - ***Machine Learning: a Probabilistic Perspective, Kevin Murphy (https://www.cs.ubc.ca/~murphyk/MLbook/) Introduction to Machine Learning, Ethem Alpaydın, 3rd edition, MIT Press, 2015.***
  - ***Bayesian Reasoning and Machine Learning, David Barber, Cambridge University Press, 2012 (online version available) http://web4.cs.ucl.ac.uk/staff/D.Barber/pmwiki/pmwiki.php?n=Brml.HomePage***





Introduction to Machine Learning

- **Grading:**
  - Homeworks/Quizzes/Polls % 20
  - Midterm exam % 30
  - Project % 10
  - Final Exam % 40

Homeworks

- Theoretical questions
- Programming assignments
  - You may discuss with your peers
  - You may not share code, pseudocode or notes
  - This also applies to solutions/code found on the internet
  - Should some submitted homework assignments be identical or suspected to be identical, all involved parties will get a grade of zero from all homeworks.
- You will make a demo of your homework and you will get a grade based on the demo and report.

Projects

- Will be done in groups of two students
- Phases
  - Topic selection and proposal
  - Midterm report
  - Final report
  - Final presentation and demo
- You should implement your own code
- Start thinking about your topic!

Web Site

- The class web page is available at: https://classroom.google.com
- Class code: oflhnjp
  - Use a gmail address containing your name and surname
  - Please click on the plus sign at top right corner to join the class using the class code
  - To access the lecture notes and homeworks click on the “Classwork” tab at the top of the page.

Attendance

- Attendance to at least 70% of the lectures in mandatory to pass the course.
- Otherwise your letter grade from the course will be DZ.

Academic Honesty

- Violations of scholastic honesty include, but are not limited to cheating, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting work of another person or work previously used without informing the instructor, or tampering with the academic work of other students.

- In case academic dishonesty is observed, the first authority is the instructor of the course. The instructor may decide to give the student zero for the homework(s)/lab(s)/exam(s), give the letter grade FF, or may take disciplinary action.